# Health Paws Viz

## AI Animal Care & Pet Health Visualization

Controlled Image Generation using Stable Diffusion

Author: Prathyu Adari

ID: 16371669

## Scenario Description

- Problem: Veterinary education and animal welfare require accurate, compassionate visual representation.
- Goal: Build a system that takes structured input (breed, condition, environment) and outputs realistic, responsible, AI-generated educational images.

# Dataset Context

- Datasets Used/Referenced: Oxford-IIIT Pet Dataset & Stanford Dogs Dataset.
- Purpose: To understand the taxonomy and visual traits of specific breeds to inform our prompt engineering and evaluate the accuracy of our generated outputs.

# Methodology & Pipeline

- Core Engine: PyTorch + Hugging Face diffusers.
- Model: Stable Diffusion v1.5.
- Process:
  - - Ingest structured Data (JSON)
  - - Map data to Prompt Templates
  - - Apply Control Mechanisms (Negative prompts, Guidance Scale)
  - - Generate & Post-process (Ethics Watermarking).

# Data-to-Prompt Strategy

- Input: {"animal": "Dog", "breed": "Golden Retriever", "condition": "Recovering from leg surgery", "environment": "Vet clinic"}
- Naive Prompting: Simple concatenation of words.
- Structured Prompting: "Educational veterinary illustration, high realism... A Golden Retriever. Medical context: Recovering... Setting: Vet Clinic... Soft lighting, clinical accuracy."

# Control Strategy

## Negative Prompts (Required Control):

- "distorted, deformed, bad anatomy, unrealistic, blurry, cartoon, graphic injuries, blood, distressed animal"

Classifier-Free Guidance: Set to 7.5 to force the model to adhere strictly to the structured prompt.

# Tools & Technologies

Python, PyTorch, Diffusers Library.

PIL (Python Imaging Library) for programmatic watermarking.

Google Colab (Compute).

# Results

## Scenario 1

- Baseline vs Improved

## Scenario 2

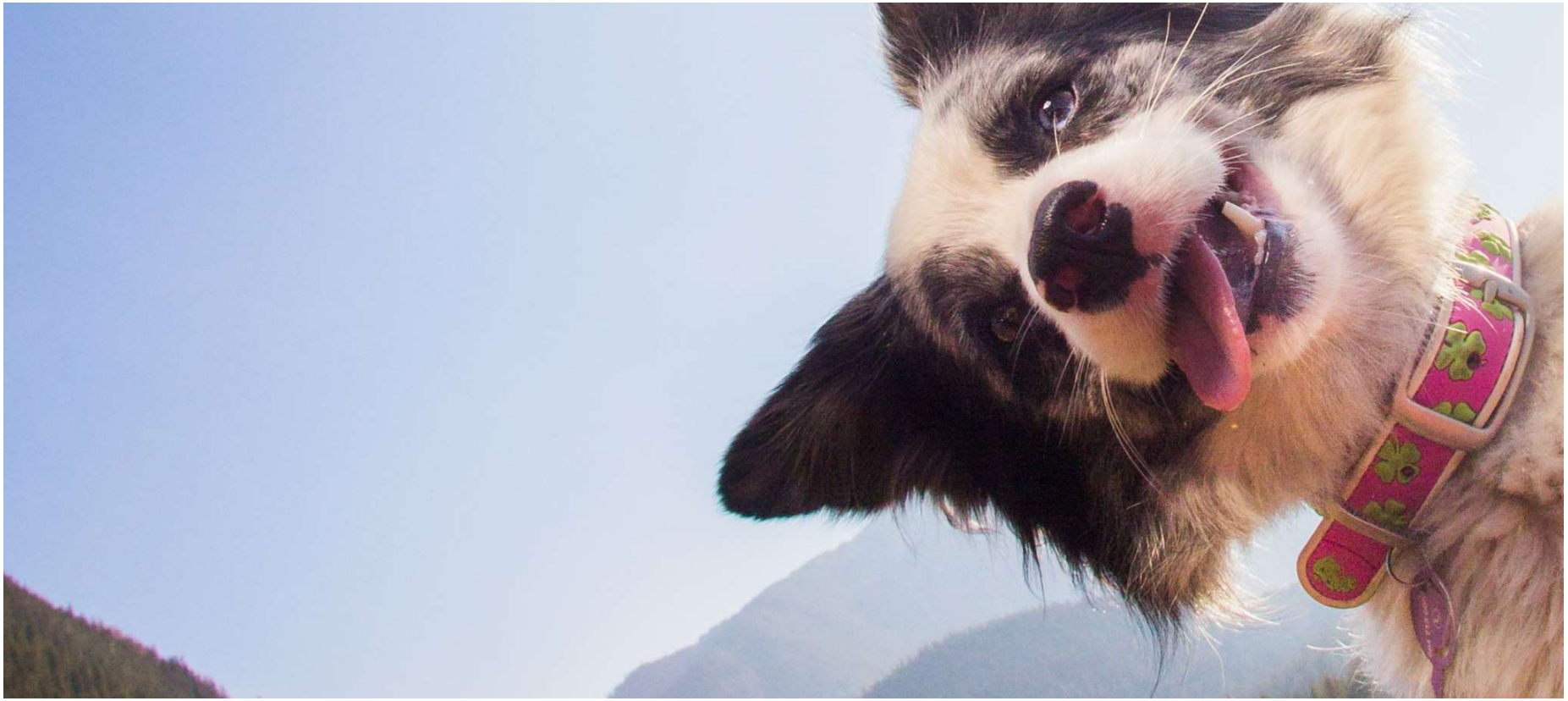
- Baseline vs Improved

# Evaluation & Failure Cases

- Metrics Applied: Prompt Alignment, Visual Quality, Ethical Adherence.
- Success: Beautifully handles standard breeds (e.g., Golden Retrievers) in clinic settings.
- Failure Case: Model struggles with complex anatomical conditions (e.g., placing a surgical cone exactly on the head without distorting the ears).

## Ethics & AI Disclosure

- Ethics: Code physically watermarks all outputs to prevent misinformation. Negative prompts explicitly ban graphic/harmful imagery.
- AI Tools Used:
  - - Gemini (Pipeline structure, README formulation, prompt engineering strategy).
  - - Copilot/ChatGPT (Code debugging and auto-completion).



# Thank you